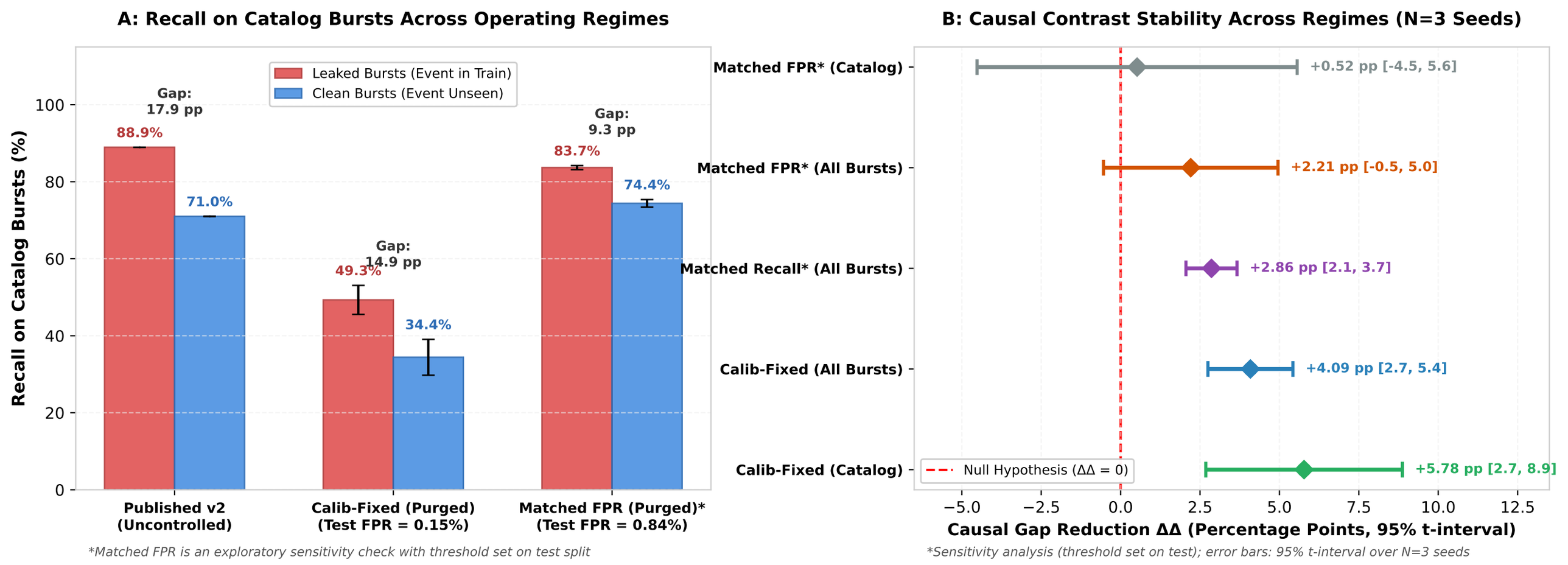
A: Recall on Catalog Bursts Across Operating Regimes
Leaked Bursts (Event in Train)
Clean Bursts (Event Unseen)
Recall on Catalog Bursts (%)
0
20
40
60
80
100
Gap:
17.9 pp
88.9%
71.0%
Gap:
14.9 pp
49.3%
34.4%
Gap:
9.3 pp
83.7%
74.4%
Published v2
(Uncontrolled)
Calib-Fixed (Purged)
(Test FPR = 0.15%)
Matched FPR (Purged)*
(Test FPR = 0.84%)
*Matched FPR is an exploratory sensitivity check with threshold set on test split
B: Causal Contrast Stability Across Regimes (N=3 Seeds)
Matched FPR* (Catalog)
+0.52 pp [-4.5, 5.6]
Matched FPR* (All Bursts)
+2.21 pp [-0.5, 5.0]
Matched Recall* (All Bursts)
+2.86 pp [2.1, 3.7]
Calib-Fixed (All Bursts)
+4.09 pp [2.7, 5.4]
Calib-Fixed (Catalog)
+5.78 pp [2.7, 8.9]
Null Hypothesis (ΔΔ = 0)
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
Causal Gap Reduction ΔΔ (Percentage Points, 95% t-interval)
*Sensitivity analysis (threshold set on test); error bars: 95% t-interval over N=3 seeds